

A sharp $C^{1,\varepsilon}$ extension of the weak anisotropic Hardy inequality

Candidate partial result generated in run `fa_banach_001`

Model GPT5.6 · agent_lane.11

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Status. Candidate substantial partial, likely valid; human review required. Under the natural normalized-regularized-distance formulation of the coefficient hypotheses, the general anisotropic conclusion holds with the exact Euclidean-distance barrier from the source theorem. No claim is made that the source paper’s literal δ -based anisotropic block assumptions are well posed on every $C^{1,\varepsilon}$ domain.

1 Source and exact open question

The source is G. Nenciu and I. Nenciu, *Weak anisotropic Hardy inequality: essential self-adjointness of drift-diffusion operators on domains in $\mathbb{R}^d$, revisited*, arXiv:2205.10494 [math.AP] (2022) [1]. Its Theorem 3.1 proves a weak anisotropic Hardy inequality on a bounded domain with C^2 boundary. In Section 5, PDF page 17, the authors ask:

“However the optimality problem for the case of rough boundaries is far from being solved. In particular an interesting open problem is whether Theorem 3.1 remains true if $\partial\Omega$ is $C^{1+\epsilon}$, $0 < \epsilon < 1$.”

For reference, the source theorem states, under its coefficient assumptions,

$$h_0[\varphi, \varphi] \geq -c_1(\nu) \|\chi_{\Omega \setminus \Gamma_\nu} \varphi\|^2 + \frac{1}{4} \int_{\Gamma_\nu} (1-q) a \delta^{\beta-2} \left[(\beta + \gamma - 1)^2 + \frac{1}{2 \log^2(1/\delta)} \right] |\varphi|^2 \rho \, dx.$$

The decisive source statement is reproduced in Figure 2.

2 Why a reformulation is necessary

Write $\delta(x) = \text{dist}(x, \partial\Omega)$. On a C^2 domain, δ is C^2 in a full boundary collar and $|\nabla\delta| = 1$. The source uses $\nabla\delta$ both in its vector field and to split the diffusion matrix into normal and tangential blocks. A $C^{1,\varepsilon}$ boundary need not have positive reach, and its cut locus can approach the boundary. Consequently δ need not be differentiable throughout any full collar. In that generality, the literal anisotropic block assumptions are not pointwise defined everywhere.

The standard remedy is a regularized distance. Ros-Oton and Serra record, for $C^{1,\alpha}$ domains, a positive function comparable to δ , of class $C^{1,\alpha}$ up to the boundary and satisfying $|D^2\psi| \leq C\delta^{\alpha-1}$ [2, Definition 2.1 and Remark 2.2]. We need the following normalized version.

the famous constant $\frac{1}{4}$ [8]:

$$\begin{aligned} \int_{\Omega} |\nabla \varphi(x)|^2 dx &\geq -c_1(\nu_1) \|\chi_{\Omega \setminus \Gamma_{\nu_1}} \varphi\|^2 + \frac{1}{4} \int_{\Gamma_{\nu_1}} \frac{1}{\delta(x)^2} |\varphi(x)|^2 dx \geq \\ &-C \|\varphi\|^2 + \frac{1}{4} \int_{\Omega} \frac{1}{\delta(x)^2} |\varphi(x)|^2 dx, \quad C < \infty \end{aligned} \quad (5.4)$$

which is known to be optimal in the sense that there exist simply connected, bounded domains with C^2 boundary for which C cannot be negative [19].

There is a large body of literature on weak weighted Hardy inequalities on domains in $\mathbb{R}^d$ with smooth/rough boundaries which can be partly traced from references in [29]. However the optimality problem for the case of rough boundaries is far from being solved. In particular an interesting open problem is whether Theorem 3.1 remains true if $\partial\Omega$ is $C^{1+\epsilon}$, $0 < \epsilon < 1$.

2. As a particular case of a more general result, for $H_0 = -\Delta$, $\rho = 1$, Ω bounded and convex G. Barbatis, S. Filippas, and A. Tertikas (see [3] and [22] for more references and

Figure 1: The open question in the original source, PDF page 17. The crop is rendered from the locally stored source PDF and retains the full page width.

3 Normalized regularized distance

Lemma 1 (normalized regularized distance). *Let $\Omega \subset \mathbb{R}^d$ be a bounded $C^{1,\varepsilon}$ domain, $0 < \varepsilon < 1$. There is a function*

$$t \in C^\infty(\Omega) \cap C^{1,\varepsilon}(\overline{\Omega}), \quad t > 0 \text{ in } \Omega, \quad t = 0 \text{ on } \partial\Omega,$$

and constants $c, C, \nu_0 > 0$ such that, whenever $\delta(x) < \nu_0$,

$$c\delta \leq t \leq C\delta, \quad t = \delta(1 + O(\delta^\varepsilon)), \quad (1)$$

$$|\nabla t|^2 = 1 + O(t^\varepsilon), \quad (2)$$

$$|D^2 t| \leq Ct^{\varepsilon-1}. \quad (3)$$

In particular, after decreasing ν_0 , $g := |\nabla t|$ is bounded above and away from zero.

Proof. Let ψ solve

$$-\Delta \psi = 1 \quad \text{in } \Omega, \quad \psi = 0 \quad \text{on } \partial\Omega.$$

The regularized-distance estimates cited above give $\psi \asymp \delta$, $\psi \in C^{1,\varepsilon}(\overline{\Omega})$, interior smoothness, and $|D^2 \psi| \leq C\delta^{\varepsilon-1}$. The Hopf lemma gives $b := |\nabla \psi| > 0$ on $\partial\Omega$, and $b \in C^\varepsilon(\partial\Omega)$.

Use the standard Whitney smoothing of the positive boundary function b to obtain a positive $B \in C^\infty(\Omega) \cap C^\varepsilon(\overline{\Omega})$, with trace b , such that near the boundary

$$|\nabla B| \leq C\delta^{\varepsilon-1}, \quad |D^2 B| \leq C\delta^{\varepsilon-2}.$$

This is the usual regularized-distance/Whitney construction; see also Fraenkel [3]. Set $t = \psi/B$, modifying B harmlessly on a compact subset of Ω so that it remains bounded away from zero. The quotient rule, $\psi = O(\delta)$, and the displayed estimates yield (3); they also give $t \in C^{1,\varepsilon}(\overline{\Omega})$. On the boundary,

$$\nabla t = \frac{\nabla \psi}{b} = n_{\text{in}},$$

$$\sup_{x \in \Gamma_{\nu_0}} |\nabla \ln(1 - q(x))| \delta(x)^s < \infty. \quad (2.21)$$

As in the case of a, r from (2.21) one has

$$\frac{1}{1 - q} \in L^\infty(\Gamma_{\nu_0}). \quad (2.22)$$

3. WEAK ANISOTROPIC HARDY INEQUALITY

Theorem 3.1. *Suppose that Assumptions A and Q hold true. Then there exists $0 < \nu_1 < \frac{\nu_0}{2}$ such that for all $0 < \nu < \nu_1$ and $\varphi \in \mathcal{C}_0^\infty(\Omega)$:*

$$\begin{aligned} h_0[\varphi, \varphi] &= \int_{\Omega} \overline{\nabla \varphi(x)} \cdot (\mathbb{D}(x) \nabla \varphi(x)) \rho(x) dx \\ &\geq -c_1(\nu) \|\chi_{\Omega \setminus \Gamma_\nu} \varphi\|^2 + \int_{\Gamma_\nu} \mathcal{H}_0(x) |\varphi(x)|^2 \rho(x) dx \end{aligned} \quad (3.1)$$

where $c_1(\nu) < \infty$, $\chi_{\Omega \setminus \Gamma_\nu}$ is the characteristic function of $\Omega \setminus \Gamma_\nu$ and

$$\mathcal{H}_0(x) = \frac{1}{4} (1 - q(x)) a(x) \delta(x)^{\beta(x)-2} \left[(\beta(x) + \gamma(x) - 1)^2 + \frac{1}{2} \left(\ln \frac{1}{\delta(x)} \right)^{-2} \right]. \quad (3.2)$$

In particular if $\varphi \in \mathcal{C}_0^\infty(\Gamma_\nu)$ then

$$h_0[\varphi, \varphi] \geq \int_{\Gamma_\nu} \mathcal{H}_0(x) |\varphi(x)|^2 \rho(x) dx. \quad (3.3)$$

Figure 2: Theorem 3.1 in the original source, PDF page 5.

the inward unit normal. Hence the Hölder continuity of ∇t gives (2).

Finally, choose a nearest boundary point y for x . Since the boundary is C^1 , $x - y = \delta(x) n_{\text{in}}(y)$. The $C^{1,\varepsilon}$ Taylor estimate at y gives

$$t(x) = \nabla t(y) \cdot (x - y) + O(|x - y|^{1+\varepsilon}) = \delta(x) + O(\delta(x)^{1+\varepsilon}),$$

which is (1). □

4 Regularized anisotropic hypotheses

Fix the t from Lemma 1, put $\Gamma_\nu^t = \{x : t(x) < \nu\}$, and set

$$e = \frac{\nabla t}{|\nabla t|}.$$

Let $\rho \in C^1(\Omega; (0, \infty))$ and let $D \in C^1(\Omega; \mathbb{R}^{d \times d})$ be symmetric and strictly positive definite. Relative to $\mathbb{R}^d = \text{span}\{e\} \oplus e^\perp$, write

$$D = \begin{pmatrix} d & d_{12} \\ d_{21} & d_{22} \end{pmatrix}, \quad d = a t^\beta, \quad q = d^{-1} d_{12} d_{22}^{-1} d_{21}.$$

Thus $0 \leq q < 1$, and the Schur complement formula gives

$$e \cdot D^{-1} e = \frac{1}{(1 - q) a t^\beta}. \quad (4)$$

Assume also

$$\rho = r t^\gamma$$

in $\Gamma_{\nu_0}^t$, where $a, r > 0$ and β, γ are C^1 . These are the direct analogues of Assumptions A and Q in the source, with t replacing δ : for some $s_\beta \leq s < 1$,

$$\begin{aligned} |\nabla \log r| + |\nabla \gamma| + |\nabla \log a| + |\nabla \log(1 - q)| &\leq Ct^{-s}, \\ |\nabla \beta| &\leq Ct^{-s_\beta}. \end{aligned} \quad (5)$$

We include explicitly the boundedness consequences used below:

$$\beta, \gamma, \log a, \log r, \log(1 - q)^{-1} \quad \text{are bounded in } \Gamma_{\nu_0}^t.$$

5 Proof intuition

The source proof uses a vector field parallel to $\nabla \delta$. Replacing δ naively by t would insert a factor $|\nabla t|^2$ into the radial derivative. Instead define

$$N = \frac{\nabla t}{|\nabla t|^2}.$$

Then $N \cdot \nabla t = 1$ exactly, so differentiating powers of t gives the same one-dimensional terms as in the smooth-distance proof. The Schur complement still supplies the factor $1 - q$. The only geometric errors come from $\operatorname{div} N$ and $|\nabla t|^{-2} - 1$. Lemma 1 bounds them by $O(t^{\varepsilon-1})$ and $O(t^\varepsilon)$. After the extra factor t in the divergence calculation, these errors are polynomially small and are absorbed by the logarithmic correction in the source barrier. A second use of the same logarithmic reserve absorbs the $O(\delta^\varepsilon)$ error in replacing t by the exact Euclidean distance δ , even in the leading k^2 term because k is bounded.

6 The anisotropic regularized-distance theorem

Theorem 2. *Assume the hypotheses above. For every $0 < \lambda < 1$ there is $\nu_\lambda > 0$ such that, for every $0 < \nu < \nu_\lambda$, there is $c_\lambda(\nu) < \infty$ for which*

$$\begin{aligned} h_0[\varphi, \varphi] &:= \int_{\Omega} \overline{\nabla \varphi} \cdot D \nabla \varphi \rho \, dx \\ &\geq -c_\lambda(\nu) \|\chi_{\Omega \setminus \Gamma_\nu^t} \varphi\|_{L^2(\rho \, dx)}^2 \\ &\quad + \frac{1}{4} \int_{\Gamma_\nu^t} (1 - q) a t^{\beta-2} \left[(\beta + \gamma - 1)^2 + \frac{\lambda}{\log^2(1/t)} \right] |\varphi|^2 \rho \, dx \end{aligned} \quad (6)$$

for all $\varphi \in C_c^\infty(\Omega)$. In particular, $\lambda = 1/2$ gives exactly the logarithmic coefficient in the barrier of the source Theorem 3.1, with t in place of δ .

More strongly, there are $\mu > 0$ and $C_\mu < \infty$ such that the exact Euclidean-distance conclusion

$$\begin{aligned} h_0[\varphi, \varphi] &\geq -C_\mu \|\chi_{\{\delta \geq \mu\}} \varphi\|_{L^2(\rho \, dx)}^2 \\ &\quad + \frac{1}{4} \int_{\{\delta < \mu\}} (1 - q) a \delta^{\beta-2} \left[(\beta + \gamma - 1)^2 + \frac{1}{2 \log^2(1/\delta)} \right] |\varphi|^2 \rho \, dx. \end{aligned} \quad (7)$$

This is the exact barrier in the source theorem for general anisotropic data, with only the coefficient hypotheses formulated relative to t .

Proof. We first recall the vector-field inequality. For every real C^1 vector field X and $\varphi \in C_c^\infty(\Omega)$, completing the square and integrating the cross term by parts gives

$$h_0[\varphi, \varphi] \geq \int_{\Omega} (\operatorname{div} X - X \cdot (\rho D)^{-1} X) |\varphi|^2 \, dx. \quad (8)$$

Put

$$k = \beta + \gamma - 1, \quad f(t) = \frac{1}{\log(1/t)}, \quad g = |\nabla t|, \quad N = \frac{\nabla t}{g^2}.$$

Then $N \cdot \nabla t = 1$, and Lemma 1 implies

$$\operatorname{div} N = O(t^{\varepsilon-1}), \quad 1 - g^{-2} = O(t^{\varepsilon}). \quad (9)$$

On a sufficiently thin collar define

$$\tilde{X} = \frac{1}{2}(1 - q)a\rho t^{\beta-1}(k + f)N.$$

Choose a C^1 cutoff Ψ , equal to one on a smaller collar and zero outside the original collar, and use $X = \Psi \tilde{X}$ in (8).

On the collar where $\Psi = 1$, let

$$P = (1 - q)a\rho t^{\beta-2}.$$

The Schur identity (4), together with $N = e/g$, gives

$$\tilde{X} \cdot (\rho D)^{-1} \tilde{X} = \frac{1}{4}P(k + f)^2 g^{-2}. \quad (10)$$

Since $tf'(t) = f(t)^2$, direct differentiation gives

$$\operatorname{div} \tilde{X} - \tilde{X} \cdot (\rho D)^{-1} \tilde{X} = P \left[\frac{1}{4}k^2 + \frac{1}{4}f^2 + E \right], \quad (11)$$

where

$$\begin{aligned} E &= \frac{1}{2}t(N \cdot \nabla k) [1 + (k + f) \log t] \\ &\quad + \frac{1}{2}t(k + f)N \cdot \nabla \log((1 - q)a\rho) + \frac{1}{2}t(k + f) \operatorname{div} N \\ &\quad + \frac{1}{4}(k + f)^2(1 - g^{-2}). \end{aligned} \quad (12)$$

For completeness, the cancellation in (11) is

$$\frac{1}{2}\{k(k + f) + f^2\} - \frac{1}{4}(k + f)^2 = \frac{1}{4}(k^2 + f^2).$$

The coefficient bounds (5), the boundedness of k , and (9) imply

$$|E| \leq C [t^{1-s}(1 + |\log t|) + t^{\varepsilon}]. \quad (13)$$

Every positive power of t , even multiplied by a fixed power of $|\log t|$, tends to zero faster than $f(t)^2$. Hence

$$E = o(f^2) \quad (t \downarrow 0)$$

uniformly along the boundary. After shrinking the collar,

$$E \geq -\frac{1-\lambda}{4}f^2.$$

Equations (11) and (8) therefore give the desired barrier on Γ_ν^t . On the compact transition region of the cutoff, $\rho^{-1}(\operatorname{div} X - X \cdot (\rho D)^{-1} X)$ is bounded below. Absorbing that bound into $c_\lambda(\nu)$ proves (6).

It remains to recover the exact source barrier. Apply (6) with a fixed $\lambda_0 \in (1/2, 1)$, say $\lambda_0 = 3/4$, and write

$$\theta = \frac{t}{\delta} = 1 + O(\delta^\varepsilon), \quad k = \beta + \gamma - 1, \quad f_\delta = \frac{1}{\log(1/\delta)}.$$

Since β and k are bounded, uniformly as $\delta \downarrow 0$,

$$\theta^{\beta-2} = 1 + O(\delta^\varepsilon), \quad f(t)^2 = f_\delta^2(1 + o(1)).$$

Consequently

$$\begin{aligned} \theta^{\beta-2} [k^2 + \lambda_0 f(t)^2] - \left[k^2 + \frac{1}{2} f_\delta^2 \right] \\ \geq -C\delta^\varepsilon + (\lambda_0 - \tfrac{1}{2} + o(1)) f_\delta^2 \geq 0 \end{aligned}$$

on a sufficiently thin collar, because $\delta^\varepsilon = o(f_\delta^2)$. Multiplying by the positive factor $(1-q)a\delta^{\beta-2}$, restricting to a smaller δ -collar contained in the t -collar, and enlarging the finite interior constant proves (7). $\square$

7 Classical Laplacian corollary

Corollary 3. *Let $\Omega \subset \mathbb{R}^d$ be a bounded $C^{1,\varepsilon}$ domain, $0 < \varepsilon < 1$, and let $\delta = \operatorname{dist}(\cdot, \partial\Omega)$. There are $\nu > 0$ and $C_\nu < \infty$ such that, for every $\varphi \in C_c^\infty(\Omega)$,*

$$\begin{aligned} \int_\Omega |\nabla \varphi|^2 \, dx \geq -C_\nu \|\chi_{\{\delta \geq \nu\}} \varphi\|_2^2 \\ + \frac{1}{4} \int_{\{\delta < \nu\}} \frac{1}{\delta^2} \left[1 + \frac{1}{2 \log^2(1/\delta)} \right] |\varphi|^2 \, dx. \end{aligned} \tag{14}$$

Thus the source question has a positive answer, with its exact barrier, in the classical case $D = I$, $\rho = 1$.

Proof. Take

$$D = I, \quad \rho = 1, \quad a = 1, \quad \beta = \gamma = q = 0.$$

Then $k = -1$, and (14) is exactly (7). $\square$

8 Verification and novelty check

The accompanying script `code/verify_algebra.py` checks symbolically the identity $tf' = f^2$, the cancellation of the mixed kf term, and the placement of the metric error $1 - g^{-2}$. It also evaluates finite asymptotic samples for the error-to-reserve ratio. These checks passed. It also checks samples

of the reserve $\delta^\varepsilon \log^2(1/\delta) \rightarrow 0$ used in the full anisotropic t -to- δ transfer. These are sanity checks, not a proof of the analytic estimates.

A bounded novelty search was performed on 23 July 2026 before promotion. The searched material comprised the run’s registry and solution indexes, arXiv/web searches for the exact quoted question, the phrases “weak anisotropic Hardy C^1 boundary” and “weak anisotropic Hardy $C^{1,\alpha}$ ”, and citation/title searches for arXiv:2205.10494. No later paper explicitly answering this question was found. The search did find the standard regularized-distance estimates of Ros-Oton and Serra [2]; that paper does not state the anisotropic theorem proved here. This was a bounded search, not a claim of exhaustive literature novelty.

9 Limitations and review recommendation

Theorem 2 is a complete theorem under the displayed t -based hypotheses, and its conclusion (7) uses the exact source barrier with δ for general anisotropic data. The remaining limitation lies only in the hypotheses: it does not identify normal/tangential blocks relative to $\nabla\delta$ at points where δ is not differentiable, nor claim an equivalence between arbitrary a.e. source δ -factorizations and the t -factorizations used here.

Human review should focus first on Lemma 1, especially the normalization by the smoothed boundary function B . The next priority is the four-term error formula (12); finally, check the uniform anisotropic t -to- δ comparison leading to (7).

References

- [1] G. Nenciu and I. Nenciu, *Weak anisotropic Hardy inequality: essential self-adjointness of drift-diffusion operators on domains in $\mathbb{R}^d$, revisited*, arXiv:2205.10494 [math.AP], 2022. <https://arxiv.org/abs/2205.10494>.
- [2] X. Ros-Oton and J. Serra, *Boundary regularity estimates for nonlocal elliptic equations in C^1 and $C^{1,\alpha}$ domains*, Annali di Matematica Pura ed Applicata **196** (2017), 1637–1668; arXiv:1512.07171. <https://arxiv.org/abs/1512.07171>.
- [3] L. E. Fraenkel, *On regularized distance and related functions*, Proceedings of the Royal Society of Edinburgh Section A **83** (1979), 115–122. <https://doi.org/10.1017/S0308210500011434>.